



Alexandra Rae Santora, M.S.

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Passionate and enthusiastic data scientist with 4+ years of experience working with complex biological next-generation sequencing (NGS) datasets. I'm proficient in R, Python, BASH/UNIX shell scripting and SQL for data analysis, and visualization tools like R markdown and matplotlib to create clear, useful reports. I have a strong background in cleaning and organizing large datasets, writing and executing statistical report plans, and troubleshooting and optimizing complex NGS workflows to improve efficiency. Experience meeting directly with a wide range of customers to turn their questions into implementable analysis plans. I love collaborating with cross-functional teams to solve problems and translate business needs into practical data solutions. Whether working independently or as part of a team, I'm always looking for ways to make data more accessible and actionable.

EDUCATION

University of Southern California

Master of Science, Environmental Microbiology
May 2022

Rider University

Division 1 Athlete
Bachelor of Science, Biology
May 2018

KEY SKILLS & EXPERIENCE

- 2.5 years experience in a CLIA/CAP regulated environment
- Experienced in working within an Agile project management framework
- Experience working with SciLife, LIMS, Salesforce for project management
- 6+ years experience scripting in R (tidyverse, data cleaning, statistical analysis)
- 3+ years experience scripting in Python
- 2+ years experience working with SQL
- 7+ years experience working with UNIX-based OS
- 8+ years of ad-hoc data exploration experience
- 6+ years experience working with a High-Performance Computing Cluster
- 2.5 years working with cloud computing (AWS: EC2, EBS, AMI, S3, FSX)
- 2.5 years meeting with customers of diverse backgrounds to translate their research goals into executable sequencing and statistical analysis plans.
- 2.5 years of generating deliverable detailed visual and statistical data analysis reports to customers of industry, clinical, and academic backgrounds.
- 2.5 years creating and executing SOPs and validation plans for wet- and dry-lab protocols.
- *Statistical Analysis:* Generalized Linear Models (GLMs), Mixed-Effects Models, Bayesian Statistics, Machine Learning, & Non-parametric Methods, time series analysis, survival analysis
- *Communication:* Excellent teamwork, scientific communication, and time management; thrives working across multiple teams to bridge gaps between R&D, science and SWE.

PROFESSIONAL EXPERIENCE

Bioinformatician II

March 2022 - Dec 2024

Diversigen, INC., Minneapolis, MN (remote)

- Worked directly with the software engineering team to develop, optimize & troubleshoot modular & scalable code for NGS bioinformatics workflows, including data cleaning, statistical analysis, & visualizations for multi-omic microbiome datasets (BASH, Python, R, nextflow).
- Translated bioinformatics analysis plans into reproducible code for descriptive statistics, statistical modeling, machine learning & data visualization (R).
- Met with customers to discuss study design, implementation, statistical analysis, & outcomes.
- Performed quality control, annotation, processing, & analysis of microbiome sequence data across multiple 'omics' platforms.
- Analyzed large-scale multi-omic datasets from various sources, including clinical trials, academic research, & public data repositories.
- Utilized version control systems (Git, GitHub, GitLab) for collaborative development of bioinformatics code & review of statistical analysis scripts.
- Validated bioinformatics pipelines & wet lab protocols, ensuring the accuracy & reliability of computational & experimental workflows.
- Managed large-scale dataset storage on AWS S3 & performed computational analysis using AWS EC2 for high-throughput bioinformatics data.
- Maintained compliance with CLIA & CAP standards in the context of bioinformatics research for clinical & academic projects.
- Developed & validated 20+ SOPs & Validation reports for new wet- & dry-lab methods.

Research Fellow

June - August 2017

NASA Jet Propulsion Lab - California Institute of Technology, Pasadena, CA

- Collected samples from the Perseverance Rover flight hardware to ensure compliance with Planetary Protection bioburden requirements.
- Characterized extreme microbes inhabiting cleanroom environments & spacecraft-associated hardware using selective isolation, cultivation methods, & MALDI-TOF spectrometry.

Graduate Research + Teaching Assistant

July 2018 – May 2022

University of Southern California, Los Angeles

Research Advisor: Jed Fuhrman

- Processed thousands of ocean samples from collection through sequencing.
- *National Geographic feature*: fluorescent microscopy image of marine viruses. February 2021.
- Developed a method to maximize DNA & RNA recovery from RNA preservation solution.
- Analyzed microbial community spatial variability using SSU rRNA at the ASV level.
- Optimized the lab's nitrite chemical assay.
- Completed PhD-level graduate coursework while obtaining a master's degree.
- Taught undergraduate labs in Organismal + Evolution & Cell Biology + Physiology.
- Contributed to the 20+ year NSF-funded microbial time series project through monthly sampling & processing.
- Managed lab tasks, including organization, cleaning, data organization, & equipment maintenance during management gaps.
- Member of the Simon's Foundation CBIOMES group.

Assistant Cross-Country & Assistant Track & Field Coach Saint Monica's Catholic High School, Santa Monica, CA

July 2020 - July 2022

- Revived a struggling athletic program, leading the team to League Championships in 2020 & 2021 from last place in previous years.
- Developed a training program for high school runners, helping them improve their 3-mile time by up to 6 minutes in one season.
- I worked 20 hours a week as a full-time graduate student & training for my own competitive racing.

RESEARCH PRESENTATIONS

- *2022 Ocean Sciences Meeting*: “Exploring differences in microbial community structure across the gulf stream front on a small spatial scale.” Rae Santora, Jed Fuhrman, Mick Follows, BB Cael, Delaney Nolin.
- *Simon’s Foundation CBIOMES June 2022 e-meeting (oral)*: “Exploring differences in microbial community structure across the gulf stream front on a small spatial scale, cont’d.” Rae Santora, Jed Fuhrman, Mick Follows, BB Cael, Delaney Nolin.
- *Simon’s Foundation CBIOMES December 2020 e-meeting (oral)*: “Exploring differences in microbial community structure across the gulf stream front on a small spatial scale.” Rae Santora, Jed Fuhrman, Mick Follows, BB Cael, Delaney Nolin.
- *2020 AtlantECO Workshop on Standard Protocols, online (oral)*: “Mock communities and universal primers: validating quantitative amplicon analyses.” Rae Santora, Jed Fuhrman, Yi-Chun Yeh, David Needham, Alma Parada, Erin Fichot.
- *2020 University of Southern California Marine Biology + Biological Oceanography Graduate Student Seminar series, online (oral)*: “Florida straits: Fine-spatial resolution microbial community assessment across the Gulf Stream front.” Alexandra Rae Santora, Delaney Nolin, BB Cael, Jesse McNichol, Mick Follows, Jed Fuhrman.
- *2019 University of Southern California Marine Biology + Biological Oceanography department retreat, Catalina Island (poster)*: “Florida straits: Fine-spatial resolution microbial community assessment across the Gulf Stream front.” Alexandra Rae Santora, Delaney Nolin, BB Cael, Jesse McNichol, Jed Fuhrman.
- *2018 Rider University Independent Scholarship and Creative Activities Awards Ceremony (oral)*: “The Effect of Temperature Stress on Caspase-Like Activity in *Haloferax volcanii*.” Alexandra Rae Santora, Shayna Holness, Kelly Bidle, September 2016.
- *2016 Emory STEM Research and Career Symposium, Atlanta, Georgia (poster)*: “The Effect of Salt Stress on Caspase-Like Activity in *Haloferax volcanii*.” Alexandra Rae Santora, Shayna Holness, Kelly Bidle, September 2016.

ATHLETICS: NCAA INTERCOLLEGIATE + POST-COLLEGIATE

Janes Elite Racing Team, Elite running team

June 2020 – May 2023

Sponsors: Tracksmith + Coros, *Santa Monica, CA*

- Competed in trail, cross country, road, and track races.
- Managed team’s Social Media accounts.
- 2020 USATF Southern California Club Cross Country individual champion.

Rider University, NCAA Division I Women’s Cross Country, Indoor, Outdoor Track & Field
Lawrenceville, NJ

July 2014 – June 2018

- *Captain*
- Managed 25-30 hours weekly for athletics, including year-round competitions and travel, alongside a full course load and laboratory research commitments.

REFERENCES AVAILABLE UPON REQUEST